

TravelX

Smart Currency Exchange Reservation Platform

Product Requirements Document

Version 1.2

FPT IS Internship Project

Author: Gyeonhui Lee, Jihun Yu, Seohyun Cha, Doohyun KIM, Chaei Shin

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1. Project Overview

1.1 Background

Travelers going abroad often have to visit multiple currency exchange offices in person to compare rates, and wait in long lines to exchange money. Most exchange offices still operate on a walk-in basis, making it difficult to reserve an exchange amount in advance.

This project aims to provide an online reservation platform that lets travelers compare exchange rates, reserve the currency they want, and pick it up at their chosen branch.

1.2 Goals

Provide a platform where users can complete the following within minutes:

- Compare exchange rates
- Reserve foreign currency
- Select a pickup location
- Receive a reservation confirmation

1.3 Key Differentiators

Redefine the core value as “benchmark-rate exchange comparison (refreshed daily, with branch-level preferential rates updated in real time) + reservation,” and add two AI-assisted features as MVP+: an optimal exchange-timing recommendation and a nearby exchange-shop recommendation, to demonstrate experience with financial domain data analysis and external API integration.

1. Daily-updated benchmark-rate comparison, with real-time branch-level preferential rates — integration with external public data APIs
 2. Online reservation and confirmation issuance — reservation-only stock and concurrency handling
 3. AI-based optimal exchange timing recommendation — time-series data analysis and batch processing
- AI-based exchange shop recommendation — rule-based scoring evolving into click/conversion-driven weight optimization

2. Target Users

2.1 Primary Users

- Overseas travelers
- Business travelers
- Walk-in exchange customers

2.2 Secondary Users

- Exchange office staff

3. Problem Definition

- It is hard for users to know which exchange office offers the best rate.
- Users must visit a branch in person to exchange currency.
- Popular currencies frequently run out of stock.
- Wait times are unpredictable.

4. Solution

- Daily-updated benchmark-rate lookup, with real-time branch preferential rates
- Compare rates across exchange offices
- Online reservation
- Select pickup date and time
- Reservation confirmation issuance
- AI-based exchange timing recommendation
- AI-based exchange shop recommendation

5. Competitive Service Comparison

TravelX's positioning compared to existing services is as follows.

Service Type	Example	Limitations
Bank app currency exchange reservation	KB Kookmin Bank, Shinhan Bank, etc.	Only offers the bank's own rate; cannot compare with other exchange offices
Airport/city exchange kiosks	Individual exchange office counters	No real-time comparison; requires waiting on site
Travel community exchange rate info	Informally shared via Naver Cafe, etc.	Not real-time; no reservation feature
TravelX (this project)	-	Provides daily-updated rate comparison across multiple exchange offices + online reservation + AI-based timing & shop recommendations, all on one platform

6. MVP Scope

6.1 User Features

Authentication

- Sign in with Google
- Log out

Exchange Rate

- View exchange rates
- Search currency
- Compare rates across exchange offices

Reservation

- Select currency
- Enter amount
- Select pickup location
- Select pickup date/time
- Confirm reservation

Reservation Management

- View reservations
- Cancel reservation
- View reservation history

Cash Pickup

- Present reservation confirmation

Notifications

- Reservation completed notification
- Reservation cancelled notification
- Pre-pickup reminder notification

6.2 Admin Features

- Manage exchange offices (branches)
- Manage currency inventory
- Approve/manage reservations
- Manage exchange rates
- Manage users
- Dashboard (reservation status, popular currency stats)

6.3 AI Exchange Timing Recommendation

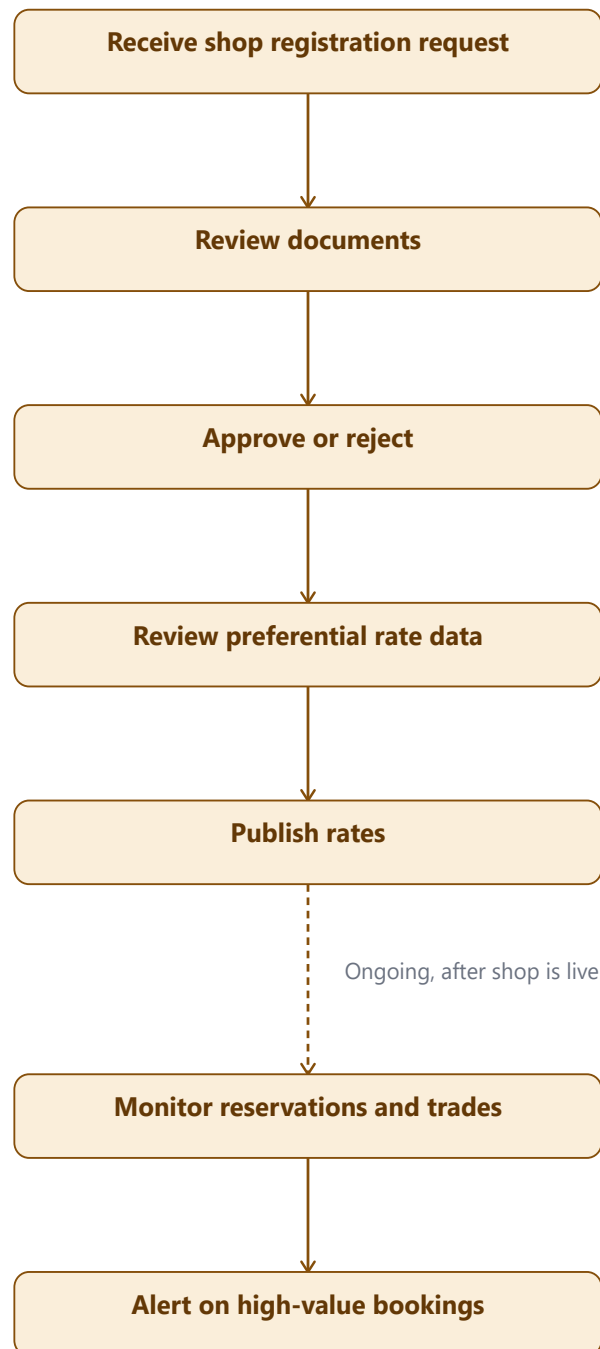
- Compares the current rate to the regression model's predicted value to generate a recommendation

6.4 AI Exchange Shop Recommendation

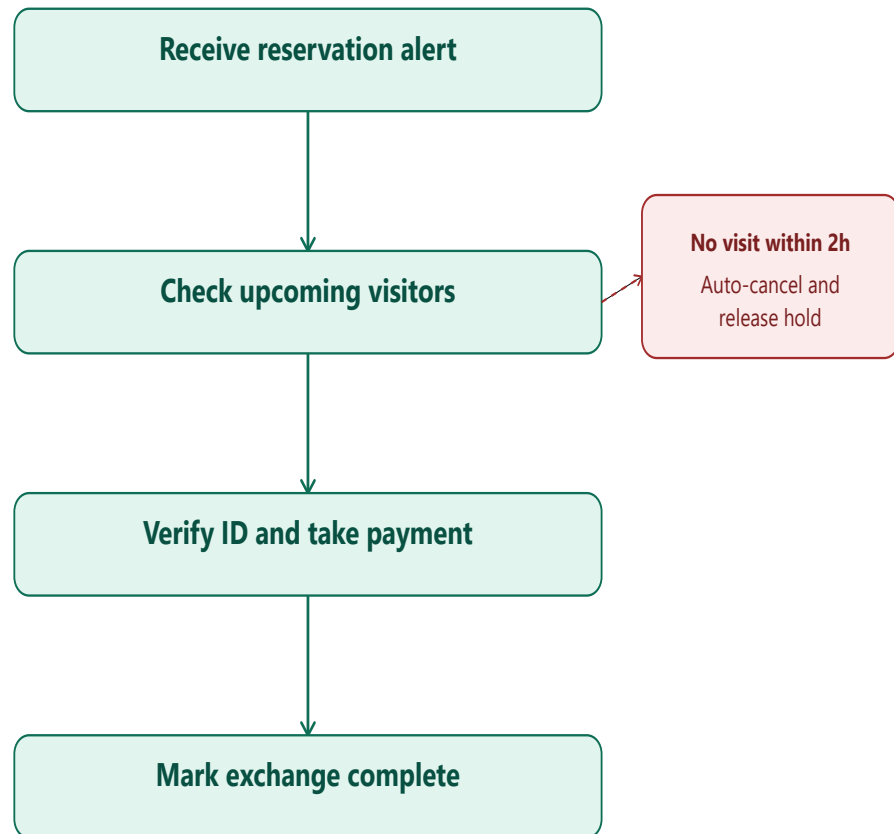
- Ranks nearby exchange shops by a distance/rate/availability/reservation composite score to help users pick among options.

7. User Flow

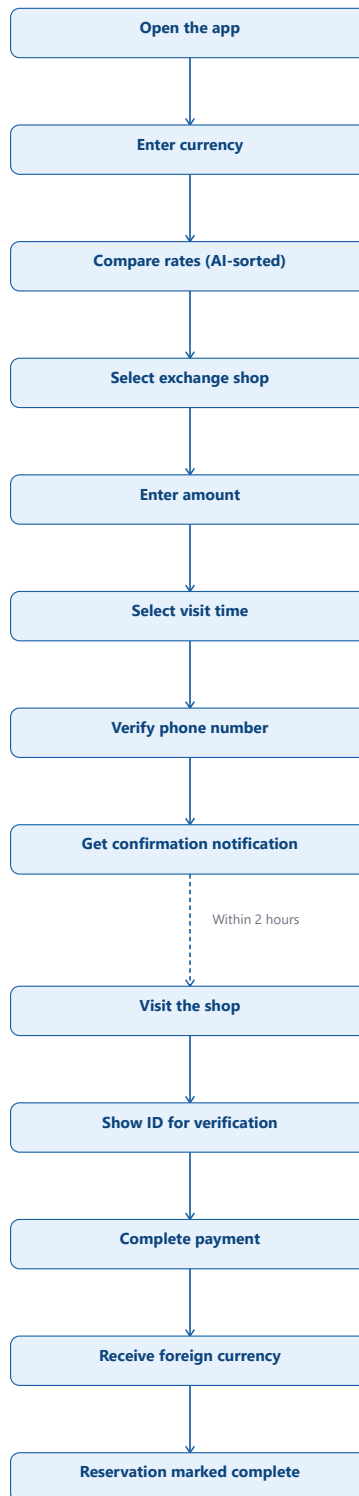
7.1 Userflow Admin



7.2 Userflow Staff



7.3 Userflow Tourist



8. Functional Requirements

Feature	Input	Process	Output
Sign In with Google	Google OAuth2 ID token	Verify token with Google; auto-create account on first login or retrieve existing user; issue JWT	User account created/retrieved, session started
Logout	JWT (session)	Invalidate/revoke JWT	Session ended
Currency Search	Currency code or country name	Query currency list	Matching currency list
Exchange Rate Comparison	Currency, (optional) location	Query benchmark rate + branch preferential rates	Ranked rate comparison list
Reservation	Currency, amount, branch, date/time	Validate reservation-only stock, lock rate, hold stock for 2 hours	Reservation created (rate & currency guaranteed)
Cancel Reservation	Reservation ID	Validate ownership & status, release held stock	Reservation cancelled, stock restored
Reservation History	User ID (session)	Query past reservations	Reservation history list
Reservation Confirmation	Reservation ID	Generate reservation number	Confirmation
Notifications	Reservation status change event	Generate in-app notification	Reservation completed/cancelled/pickup-reminder notification
Rate Management (Admin)	Currency, new benchmark rate	Validate & update rate	Exchange rate updated
Branch Management (Admin)	Branch details	Create/update/deactivate branch	Branch record updated
Reservation Management (Admin)	Reservation ID, action	Approve/modify/cancel reservation	Reservation status updated
User Management (Admin)	User ID, action	View/suspend/update user account	User account updated
AI Exchange Timing	Recent N-day rate	Run regression model, compare prediction to current rate	Now/Wait/Neutral badge

Feature	Input	Process	Output
Recommendation	history		
Admin Dashboard	Date range filter	Aggregate reservation & currency stats	Reservation status & popular currency statistics
AI Exchange Shop Recommendation	User location, currency, radius, top_n	Score branches (distance/rate/availability/reservation) & rank	Ranked nearby branch list
Identity Verification (KYC)	Phone number (booking) / ID document (pickup)	Verify phone at reservation, verify ID at pickup	Verified reservation / verified pickup
No-Show Trust Control	User's no-show history	Check 30-day no-show count against thresholds	Reservation limit applied (1 concurrent / 7-day block)
Large-Amount Reservation Limit	Reservation amount	Validate against USD 10,000 threshold	Reservation blocked/flagged if at or above threshold
Branch Rate & Stock Setting	Preferential rate (%), reservation-only stock quantity	Branch submits daily values via admin page	BranchCurrencyRate updated

9. Detailed Design of the AI Exchange Timing Recommendation Feature

9.1 Purpose

To help travelers who find it hard to judge whether to exchange currency now or wait, this feature provides a reference recommendation signal based on exchange rate history data.

9.2 Data Sources

- Exchange rate history loaded daily into our own DB
- Exchange rate data collected in batches from external public APIs (e.g., the Korea Eximbank Open API)

9.3 Recommendation Logic — Phase 1

Instead of complex time-series models like ARIMA or LSTM, the design uses a Linear Regression model with moving averages and volatility as features. This offers a step up in modeling capability from a simple moving-average rule model, while remaining a low-risk structure that can be reliably implemented and validated within a month.

1. Retrieve exchange rate data for the most recent N days
2. Build features: recent moving average and standard deviation, recent rate of change, lagged exchange rate values, etc.
3. Using the next day exchange rate as the target, train a linear regression model
4. Compare the predicted value with the current rate to produce a "Exchange Now"/"Wait"/"Neutral" signal
5. Store the regression coefficients as well, so it is possible to explain which features influenced the signal and by how much.

9.4 Safeguards and Operational Considerations

- Fallback logic: if model prediction fails or input data is abnormal, automatically switch to the existing moving average $\pm$ standard deviation rule-based logic
- Warm-up policy: until at least 14 days of exchange-rate history are available for a currency, suppress the Now/Wait/Neutral badge and display a "Collecting data" state instead.

9.5 Future Enhancements — Phase 2

- Introduce time-series forecasting models
- Refine predictions by combining economic indicators and news sentiment analysis
- Set per-user target exchange rate alert thresholds
- Expand to multiple currency pairs and enhance the automatic model retraining pipeline

9.6 Display and Disclaimer

Display a recommendation badge on the exchange rate comparison screen, with the following

disclaimer at the bottom: “This recommendation is not investment advice and is provided for reference only.”

10. Non-functional Requirements

Category	Requirement
Response time	Average within 2 seconds
Availability	99% or higher
Security	JWT authentication (Google OAuth2 login only), role-based authorization
Scalability	REST API, Layered Architecture
Batch stability	Falls back to the latest cached data if the external exchange rate API fails, with retry logic applied
Data consistency	Transaction/lock handling to prevent inventory from being oversold during concurrent reservations

11. System Architecture Overview

The system is based on a 3-tier structure: Frontend (React/TypeScript) → Backend REST API (Spring Boot) → MySQL database, with a separate batch scheduler configured to collect exchange rate history.

- Frontend: React, TypeScript, Tailwind
- Backend: Spring Boot REST API, Spring Security, JPA/QueryDSL
- AI: Python, scikit-learn, PyTorch, pandas
- Scheduler: Spring Scheduler-based exchange rate batch collection (runs daily)
- Database: MySQL
- External integration: public exchange rate data API

12. Database Design (Draft)

User

Field	Description
id	PK
name	Name (from Google profile)
email	Email (from Google profile)
googleId	Google account unique ID (OAuth2 "sub" claim)
role	Role (USER / ADMIN)

Branch

Field	Description
id	PK
name	Branch name
address	Address
phone	Contact number
latitude	Branch latitude
longitude	Branch longitude
businessHours	Business hours

Currency

Field	Description
id	PK
code	Currency code (USD, JPY, etc.)
country	Country
buyRate	Buy rate
sellRate	Sell rate
updatedAt	Rate update timestamp

Note: buyRate/sellRate above represent the daily benchmark rate sourced from the external exchange-rate API. Branch-level final rates are derived — see BranchCurrencyRate below.

BranchCurrencyRate

Field	Description
id	PK
branchId	FK → Branch

Field	Description
currencyId	FK → Currency
preferentialRate	Branch preferential rate (%), set by the branch and applied instantly; final rate = benchmark rate × (1 – preferentialRate)
reservationOnlyStock	Quantity the branch allocates for app reservations only (separate from walk-in inventory); reservations are validated and decremented against this field
updatedAt	Last updated timestamp

ExchangeRateHistory

Field	Description
id	PK
currencyId	FK → Currency
rate	Exchange rate at time of recording
recordedAt	Recorded timestamp

Reservation

Field	Description
id	PK
userId	FK → User
currencyId	FK → Currency
branchId	FK → Branch
amount	Reservation amount
pickupDate	Pickup date
pickupTime	Pickup time
status	Reservation status

Note: reservation amount is validated against BranchCurrencyRate.reservationOnlyStock (not total branch inventory) and capped at USD 10,000 equivalent per transaction.

ReservationStatus (Enum)

- RESERVED
- COMPLETED
- CANCELLED

Notification

Field	Description
id	PK

Field	Description
userId	FK → User
type	Notification type
message	Notification content
isRead	Read status
createdAt	Created timestamp

13. Tech Stack

Area	Technology
Frontend	React, TypeScript, Tailwind CSS
Backend	Spring Boot, Spring Security, JPA, QueryDSL
Database	MySQL
AI	Python, scikit-learn, PyTorch, pandas
Infra	Kubernetes, AWS EC2
Batch/Analysis	Spring Scheduler, moving average/standard deviation statistical logic

14. API Specification (Draft)

Method	Endpoint	Description
GET	/currencies	Retrieve full list of currencies/rates
GET	/currencies/{code}	Retrieve details for a specific currency
GET	/currencies/{code}/history	Retrieve exchange rate history
GET	/currencies/{code}/recommendation	Retrieve exchange timing recommendation
GET	/branches	Retrieve branch list
POST	/reservations	Create reservation
GET	/reservations	Retrieve my reservation list
DELETE	/reservations/{id}	Cancel reservation
GET	/admin/reservations	(Admin) Retrieve all reservations
PATCH	/admin/rates	(Admin) Update exchange rates
GET	/admin/dashboard	(New, Admin) Retrieve statistics dashboard
GET	/branches/recommend	(New) Retrieve AI-ranked nearby exchange-shop recommendations (distance/rate/availability/reservation scoring)
PATCH	/branches/{id}/rate	(New, Branch) Update branch preferential rate and reservation-only stock
POST	/auth/verify-phone	(New) Send/confirm phone verification code (KYC stage 1)
GET	/oauth2/authorization/google	Initiate Google OAuth2 sign-in
GET	/auth/google/callback	Google OAuth2 callback — verify token, create/retrieve user, issue JWT
POST	/auth/logout	Invalidate JWT / end session

15. Development Schedule (4-Week Roadmap)

Week	Goal	Detailed Tasks
Week 1	Design & foundation setup	ERD design, project setup, Google OAuth2 + JWT authentication implementation; apply for exchange-rate API auth key
Week 2	Core feature implementation	Exchange rate lookup/comparison API, reservation CRUD, branch management; branch preferential-rate & reservation-only-stock management API
Week 3	Differentiating feature implementation	Notifications, exchange rate history batch collection, AI recommendation logic ; pickup-reminder batch job; AI exchange-shop recommendation scoring service
Week 4	Finalization & deployment	Admin dashboard, testing, Docker/AWS deployment, documentation

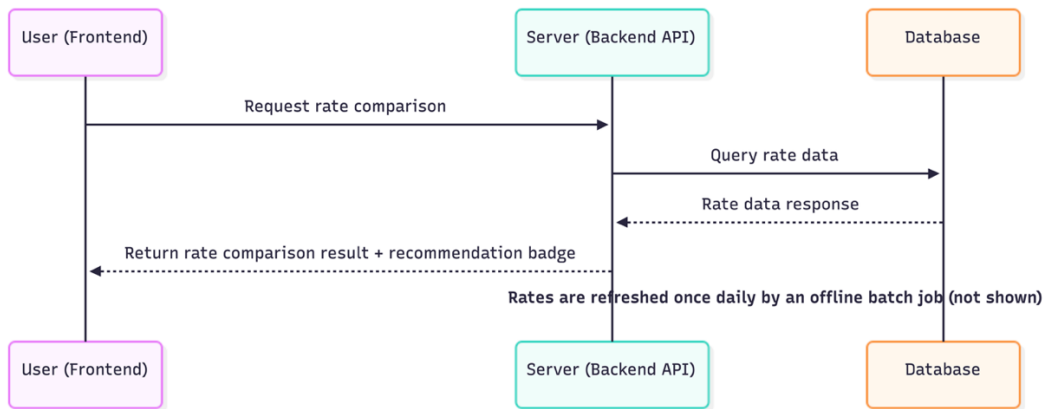
16. Future Features

- Exchange rate alerts (notify when target rate is reached)
- Google Maps-based nearby branch recommendations
- Multi-language support
- Online payment integration
- Digital wallet integration
- Enhanced AI recommendation model (time-series forecasting, news sentiment analysis)
- Evolve exchange-shop recommendation from rule-based scoring to Learning-to-Rank / contextual-bandit models as click and conversion data accumulate

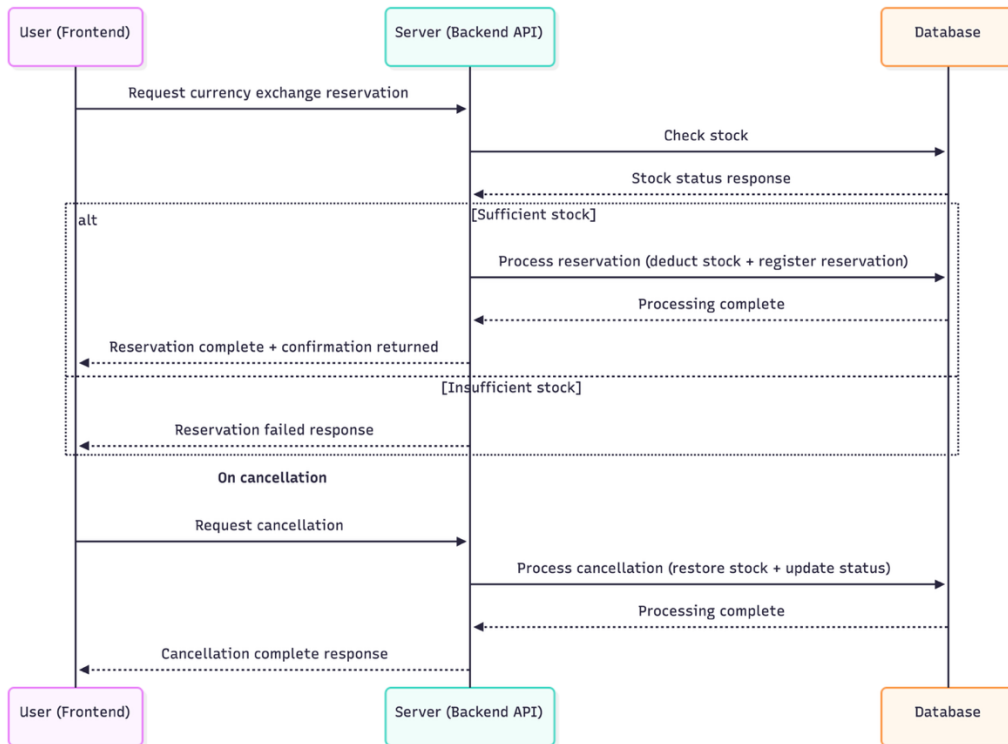
17. Sequence Diagrams

These are the exact Mermaid sources for the two core request flows, abstracted to three lanes only — User → Server → Database. Notification writes are folded into the Server→Database step rather than shown as a separate participant. Diagram images are to be inserted separately from the Mermaid source below.

17.1 Exchange Rate Comparison



17.2 Currency Exchange Reservation



18. Detailed Design of the AI Exchange Shop Recommendation Feature

18.1 Purpose

To help travelers choose among nearby exchange shops when several are available, this feature ranks candidate branches with a composite score based on distance, rate competitiveness, availability, and reservation feasibility. It complements, and is independent from, the exchange-timing recommendation in §9 — this feature ranks shops, not timing.

18.2 Inputs / Outputs

Field	Description
latitude, longitude	User's current location (required)
currency	Desired currency (required)
radius_km	Search radius (optional, default 5 km)
top_n	Maximum number of branches to return (optional, default 10)
total_score	Final recommendation score (0–1)
breakdown	Per-factor contribution: distance_score, rate_score, availability_score, reservation_score
is_open_now, reservation_available	Status flags used for UI badges

18.3 Scoring Logic — Phase 1 (Rule-based)

score = $w_1 \cdot \text{distance_score} + w_2 \cdot \text{fee_score} + w_3 \cdot \text{availability_score} + w_4 \cdot \text{reservation_score}$
 Initial weights: $w_1 = 0.35$ (distance), $w_2 = 0.35$ (rate/fee competitiveness), $w_3 = 0.20$ (availability), $w_4 = 0.10$ (reservation feasibility). Weights are configurable service-policy values, not hard-coded. Results are sorted by total_score descending, with distance_score as the tiebreaker; the top-scoring branches are highlighted on the map.

18.4 Weight Optimization — Phase 2 (Logistic Regression)

Once click/conversion data accumulates, weights w_1 – w_4 are re-derived by training a logistic-regression model that predicts click likelihood from the four sub-scores as features — the learned coefficients become the new weights, preserving the interpretable rule-based structure. Retraining runs as a periodic batch job (e.g., Spring Scheduler, daily or weekly).

18.5 Considerations

- Selection/position bias: users disproportionately see top-ranked branches; add exposure position as a feature, or apply Inverse Propensity Scoring correction.
- Conversion vs. click: to optimize for actual reservations rather than clicks, switch the training label to conversion, or use a two-stage click × conversion model.

- Phased rollout: launch MVP with rule-based scoring (Phase 1); move to logistic regression once sufficient click/conversion data exists; consider Learning-to-Rank or contextual bandits as usage matures.

18.6 Display and Disclaimer

Recommendation results are displayed as a sorted list with map highlights; the final choice of exchange shop remains with the user. The same disclaimer as §9.6 applies: this is not investment or financial advice and is provided for reference only.

19. Reservation, Cancellation & No-Show Policy

19.1 Reservation Policy

Item	MVP Policy	Rationale
Deposit	None	Minimizes friction and operational overhead
Reservation hold	2 hours from booking time	Benchmarked against bank internet-exchange practice
Cancellation	Free before visit	No penalty since no deposit is collected
Stock handling	Stock is held at booking time and auto-released on expiry	Prevents wasted inventory
Large-amount reservations	Not supported at or above USD 10,000	Avoids complex legal/reporting procedures

19.2 No-Show Risk Controls

No deposit is collected from travelers, to keep the barrier to entry low. Instead, the system protects each branch's opportunity cost through the following controls:

- User trust penalty: a user with a no-show in the last 30 days is limited to 1 concurrent reservation; 3 or more cumulative no-shows blocks new reservations for 7 days.
- Dynamic inventory rotation: a branch/time-slot's reservation-only allocation flexes with its historical no-show rate; if a user neither arrives nor can be reached as the hold nears expiry (e.g., 30 minutes before), the hold is released back to walk-in inventory.

20. Exchange Shop Partner Policy

20.1 Reservation-Only Stock Model

Branch staff cannot be relied on to keep app inventory synced with real-time walk-in sales, so TravelX avoids real-time sync entirely. Each morning, a branch sets a reservation-only quantity per currency in the admin page (e.g., 2,000 of the day's 5,000 USD on hand). Only that quantity is consumed by app bookings; the rest is sold freely walk-in and never touches app stock. Once the reservation-only quantity is exhausted, the app shows that currency/branch as fully booked for the day — an expected "sold out" state, not a bug. Branches can tune this quantity to control how much of their inventory they expose to the app

20.2 Happy-Hour Incentive

Branches may designate specific low-traffic time windows (e.g., weekday mornings) with an extra preferential rate to attract app bookings during slow periods, improving counter utilization.

20.3 Inventory & Demand Analytics Report

Branches receive aggregated reservation-trend data (by day of week, time of day, currency) to help them plan appropriate stock levels and avoid exchange-rate exposure risk from over- or under-stocking.

20.4 Non-Fulfillment Compensation & Shop Penalty Tiers

Situation	User-Facing Response	Branch Penalty
Reservation stock unavailable at pickup (branch's fault)	Immediate notification + automatic referral to the nearest branch with availability	1st occurrence: warning notice only
3 cumulative violations	Same as above	Search-ranking demotion
5 cumulative violations	Same as above	Reservation-only stock listing suspended pending re-approval

Compensation is a re-routing guarantee, not a refund, since no deposit was collected. Because penalties directly affect a branch's visibility in the app, branches that keep the reservation-only stock correctly separated are rewarded with more bookings.

20.5 Why Branches Participate

- Customer-acquisition channel for shops without their own marketing reach
- Demand forecasting from reservation data to plan staffing and inventory
- Controllable inventory-exposure risk — branches expose only as much stock as they choose

21. Legal & Compliance

21.1 Exchange Business Registration

Under Article 8(3) of the Foreign Exchange Transactions Act, operating a currency exchange business requires registration with the relevant Customs office. Registration is not required during this internship/toy-project phase; this PRD notes it as a requirement for any transition to a live commercial service.

21.2 Transaction Limits & Reporting

- ID verification is required for exchanges over KRW 1,000,000.
- Same-day exchanges of USD 10,000-equivalent or more are automatically reported to the National Tax Service; travelers carrying more than USD 10,000 out of the country must declare it to Customs. Violations: up to 5% of the amount for amounts ≤ USD 30,000, or imprisonment up to 1 year / a fine up to KRW 100,000,000 for amounts > USD 30,000.

- MVP applies this by disallowing reservations at or above USD 10,000, with an upfront notice for amounts approaching the threshold.

21.3 KYC / CDD

Exchange businesses are subject to customer due diligence under the Act on Reporting and Using Specified Financial Transaction Information for cash transactions around USD 10,000 conducted without an account (verifying identity, address, and contact information; enhanced due diligence for high-risk customers). TravelX applies a two-stage verification: phone verification at reservation time, and staff ID check at pickup — with an on-screen reminder to bring an ID for larger reservations.

22. Key Performance Indicators (KPI)

Goal: enable travelers to compare and reserve real, trustworthy branch exchange rates without waiting, while giving branches efficient inventory management and maximized offline customer acquisition.

KPI	Definition	Target
Rate-comparison usage rate	Share of all users who use the rate-comparison page	≥ 60%
Look-to-book conversion	Share of rate-detail views that start the reservation flow	≥ 20%
Reservation completion rate	Share of started reservations that are confirmed	≥ 60%
End-to-end funnel completion	Share of app entries that reach final in-branch pickup	≥ 50%
No-show rate	Share of confirmed reservations cancelled by non-visit	≤ 30%
Voluntary cancellation rate	Share of no-visit reservations cancelled voluntarily before the 2-hour expiry	≥ 50%
Branch fulfillment rate	Share of confirmed reservations completed normally with no branch-caused failure	≥ 98%
Alternate-match success rate	Share of branch-caused failures successfully re-routed to a nearby branch and picked up	≥ 40%
Average transaction value (ATV)	Average amount per completed reservation	Ongoing monitoring

23. Competitive Landscape (Updated)

23.1 Rate Comparison & Reservation

mibank is the closest direct competitor: it already compares rates across banks and private exchange shops in real time with location-based nearby-branch search, bundled with travel insurance and deposit-rate comparison. User reviews report weak support for shortchanging/counterfeit-note incidents. Because mibank has no reservation or stock-hold feature, TravelX's differentiator is the compare + reserve combination, not comparison alone. Travel Wallet is a prepaid FX card with an auto-charge "reserve at target rate" feature, but it settles entirely in-app with no pickup step, so its "reservation" has no expiry or no-show concept — reinforcing the need for TravelX's own expiry and no-show policy. Bank internet-exchange offers up to 90% preferential rates, but pickup branch/time availability varies by bank, and the Korea Federation of Banks' rate-comparison disclosure excludes private exchange shops entirely. Travel-card fee and re-exchange-fee structures across 4 major products were reviewed as a reference point for reservation-cancellation penalty design.

23.2 AI Recommendation

Coeur16's AI daily FX-prediction service reports roughly 66–70% directional accuracy at 5 business days out, even at industry-leading maturity — confirming that "90% prediction accuracy" is not a realistic MVP target. §9 and §18 use conversion/click-through KPIs instead of raw prediction accuracy. Existing travel-industry AI features (Hana Tour, Interpark Triple, Chamjoeun Travel) are all assistive recommendation layers, not autonomous decision-makers — consistent with positioning both of TravelX's AI features as reference/assistive only.

24. Wireframe Requirements (Reference)

Screen	Required Elements
Home	Currency selector, amount input, entry point to the rate-comparison list
Exchange Comparison	Per-branch rate/distance/hours list, AI-recommendation sort toggle
Reservation	Planned visit-time picker, reservation summary, expiry notice
AI Recommendation	Plain-language reason summary, recommendation ranking display
My Reservations	Reservation status (pending/complete/cancelled), countdown to expiry
Staff Confirmation (branch)	Reservation-holder info, ID-check button, payment-complete button

25. Appendix — References

- Korea Eximbank Open API — data.go.kr
- ExchangeRate-API — exchangerate-api.com

- ExchangeRate-API free tier updates once daily and does not include historical data (historical data requires a paid plan); used only as a fallback/secondary rate source.
- Google Maps Platform – Places API (Nearby Search) — reference for a future enhancement of branch-proximity search; not required for the MVP rule-based scoring in §18.